VISUAL STUDIO CODE DEVELOPMENT WITH AI INTEGRATION

Time: April 2026

Author: Huy Thanh Le

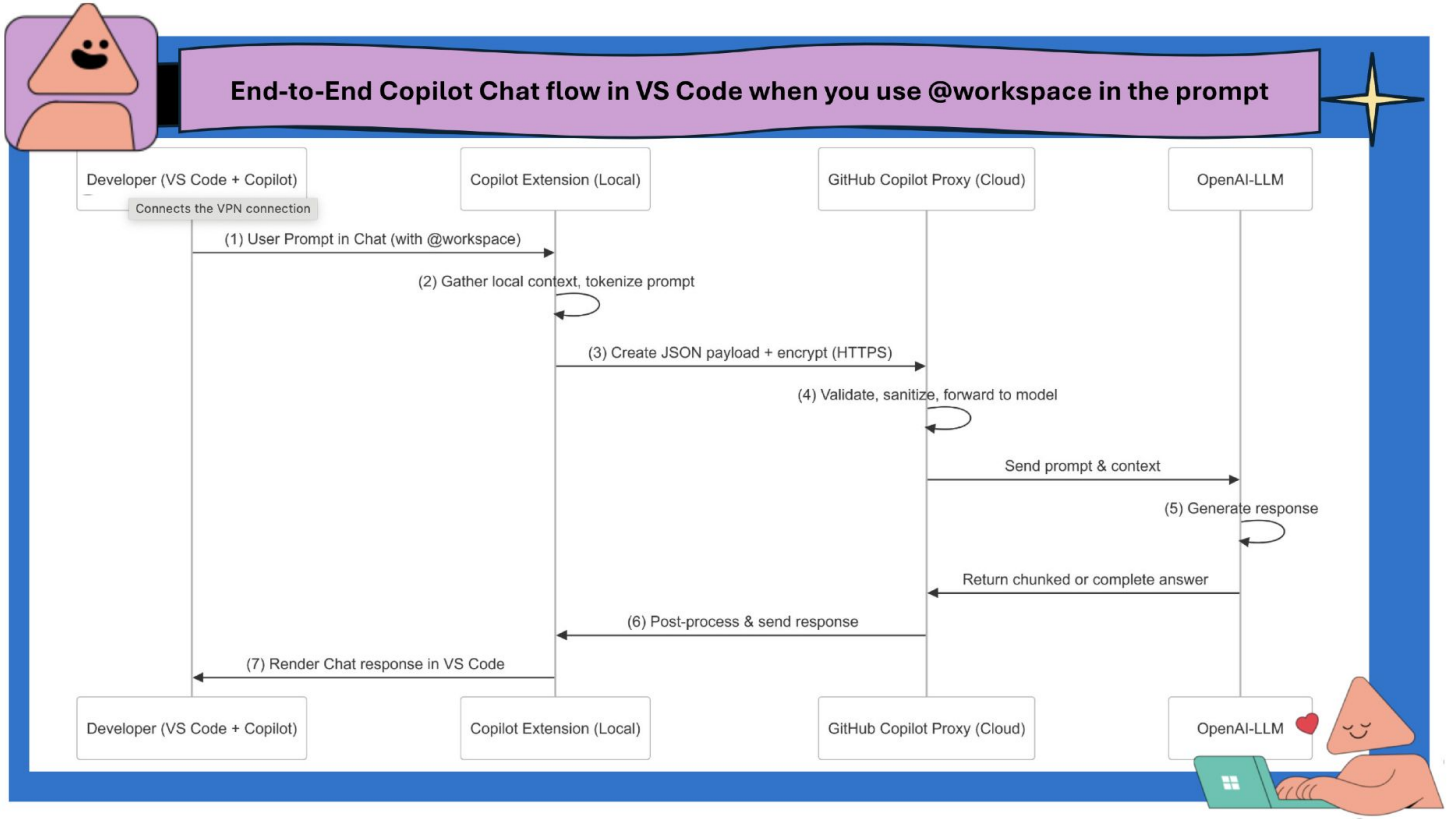
AGENDA

1. Recap Session 1 - Homework
2. Advanced Prompt Engineering Techniques
3. Building Application Session 2
4. Real-world Application Jabil
5. Summary and Recape Session 2

GitHub Copilot

Recap Session 1 - Homework

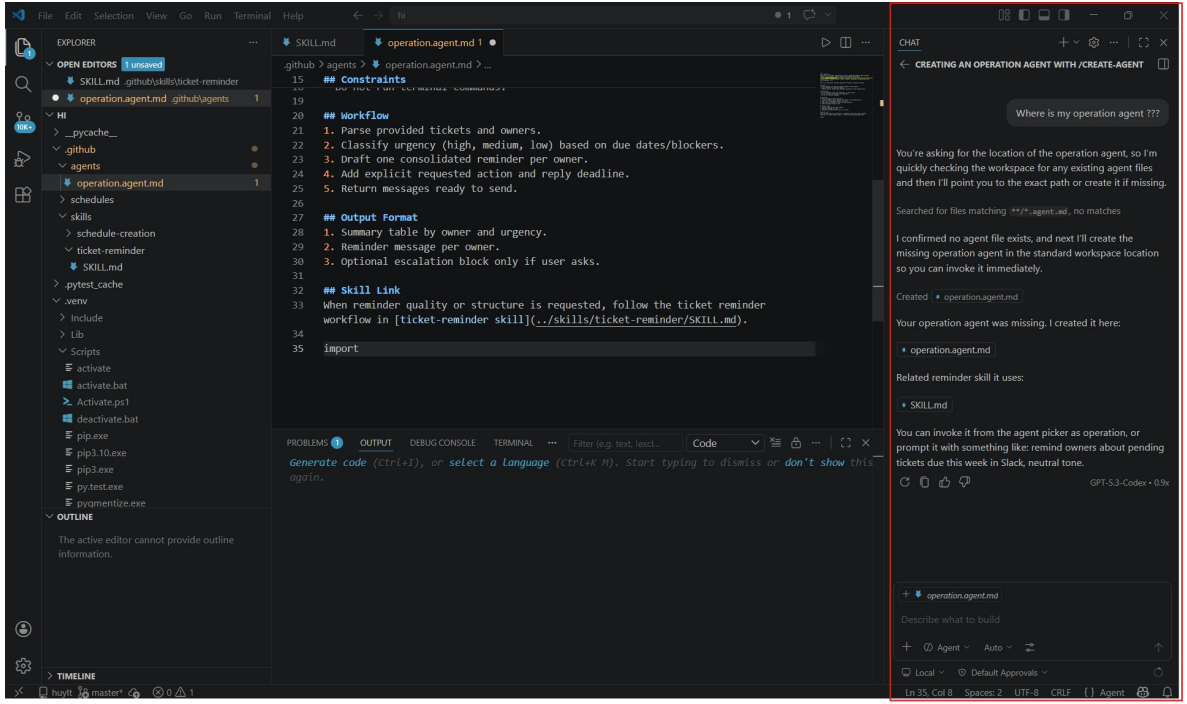
HIGHLIGHT in Session 1



Basic Copilot Features

- **Chat UI**

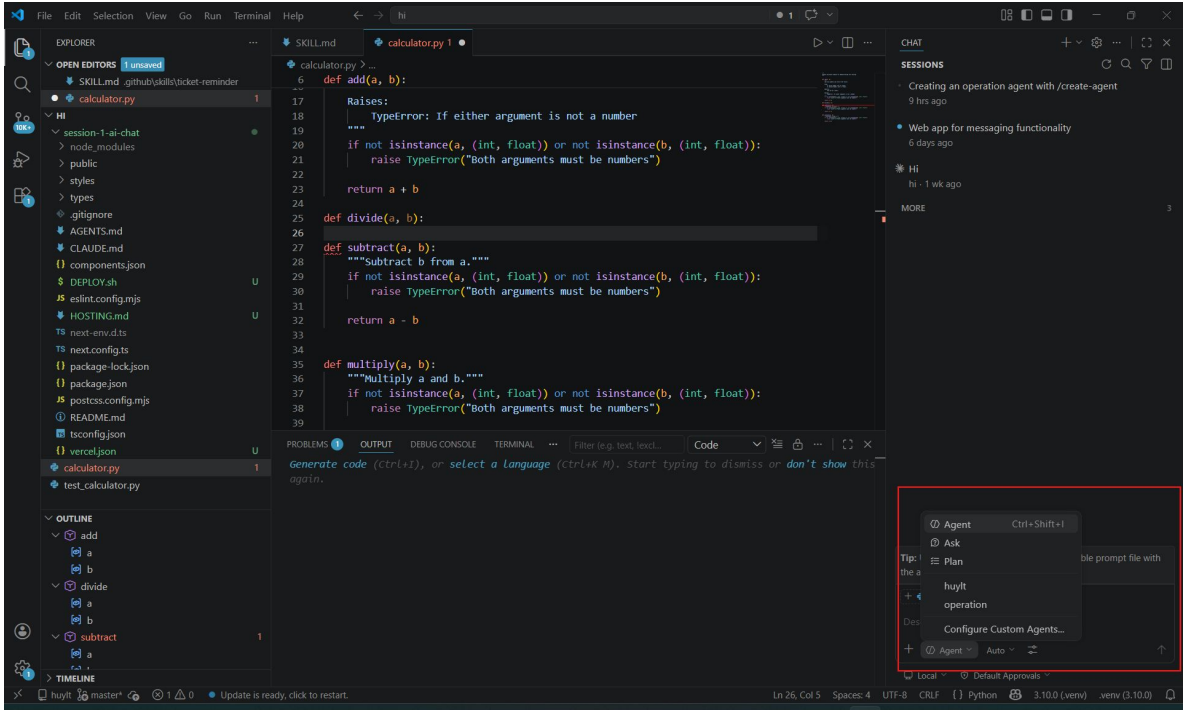
  - Natural interaction with AI Copilot
  - Quick search and information retrieval
  - AI-powered suggestions and assistance
  - Generate and refine content instantly
  - Real-time insights and responses



Basic Copilot Features

- **Plan - Agent - Ask Mode**

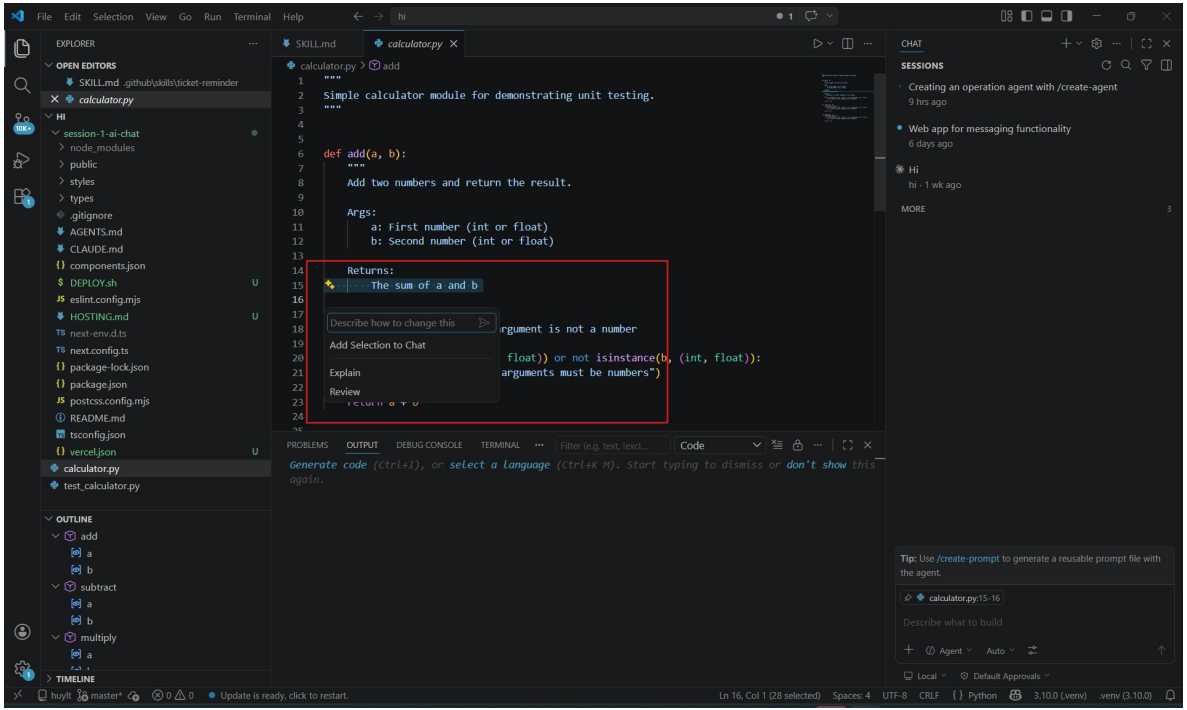
  - Plan: Break down tasks into clear steps
  - Agent: Execute tasks autonomously
  - Ask: Get instant answers and guidance
  - Seamless switching between modes
  - Improve productivity and workflow efficiency



Basic Copilot Features

- **Code Explanation**

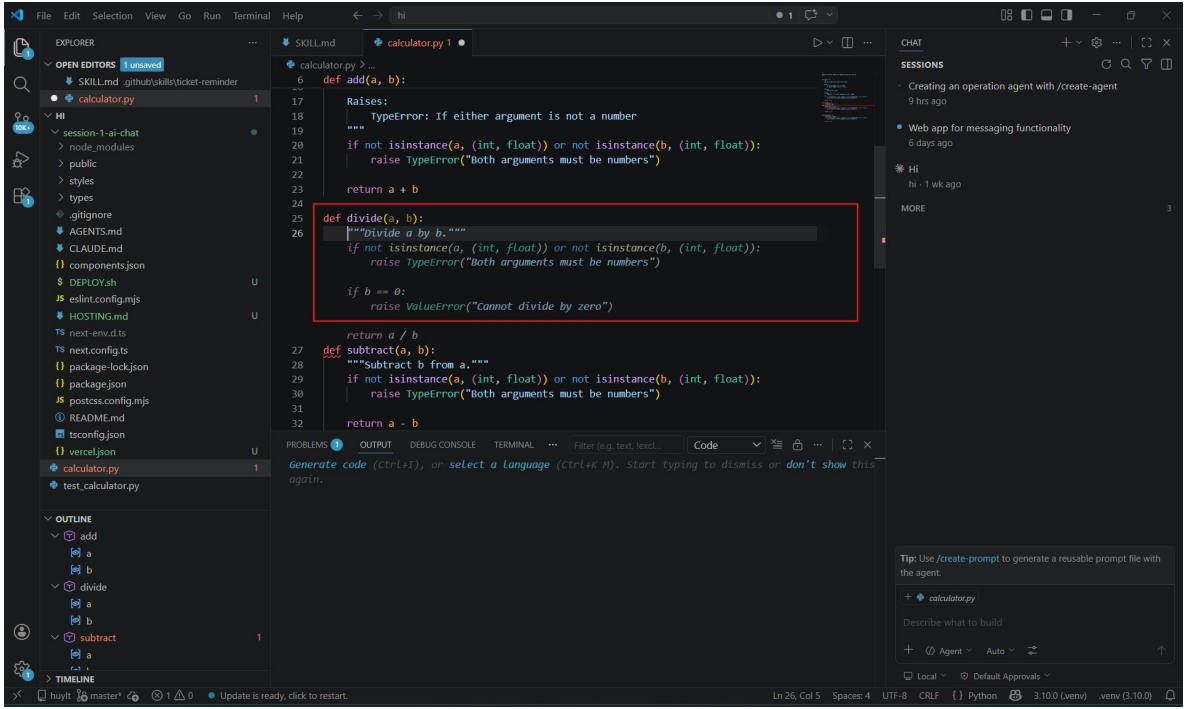
  - Explain code in natural language
  - Break down logic and functionality
  - Highlight key components and flow
  - Suggest improvements and optimizations
  - Help understand unfamiliar code quickly



Basic Copilot Features

- **In-line Suggestions**

  - Real-time code suggestions while typing
  - Auto-complete functions and logic
  - Improve code quality and consistency
  - Suggest fixes and optimizations
  - Reduce development time and errors



Quizz Session 1

Advanced Prompt Engineering Techniques

Zero - One - Few Shoot Prompt

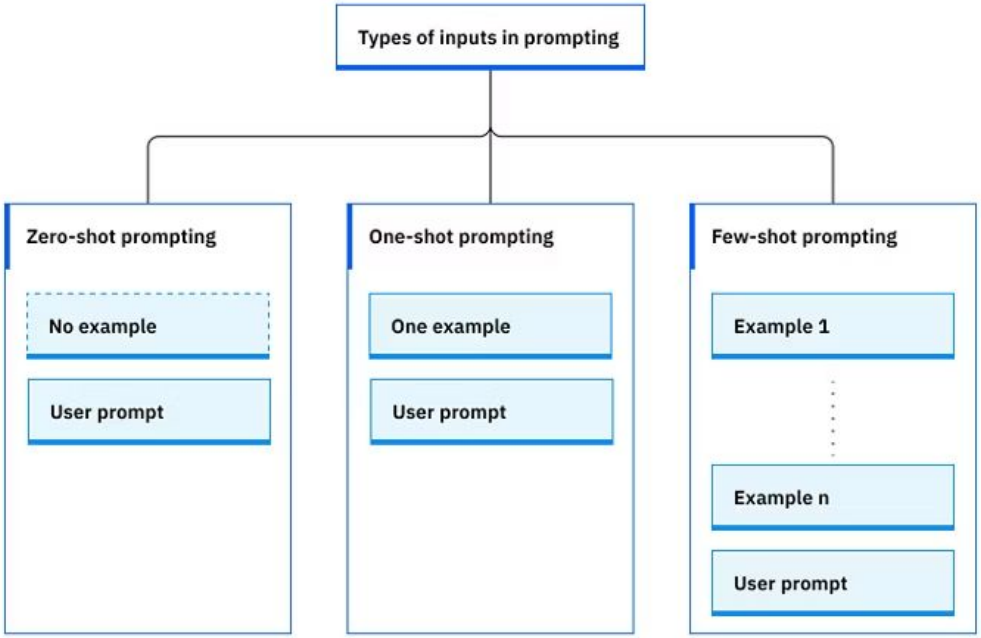
- **Zero-shot prompting**

Shows how a model takes a prompt → uses pretrained knowledge → outputs result.

No examples provided This works because LLMs rely on prior training data

- **Few-shot / One-shot prompting**

Demonstrates input-output examples before the actual task. Helps model learn pattern via ***"in-context learning"***



Architectural Pattern Prompting

Most people write prompts like:

"Refine this."

"Summarize this."

"Fix my writing."

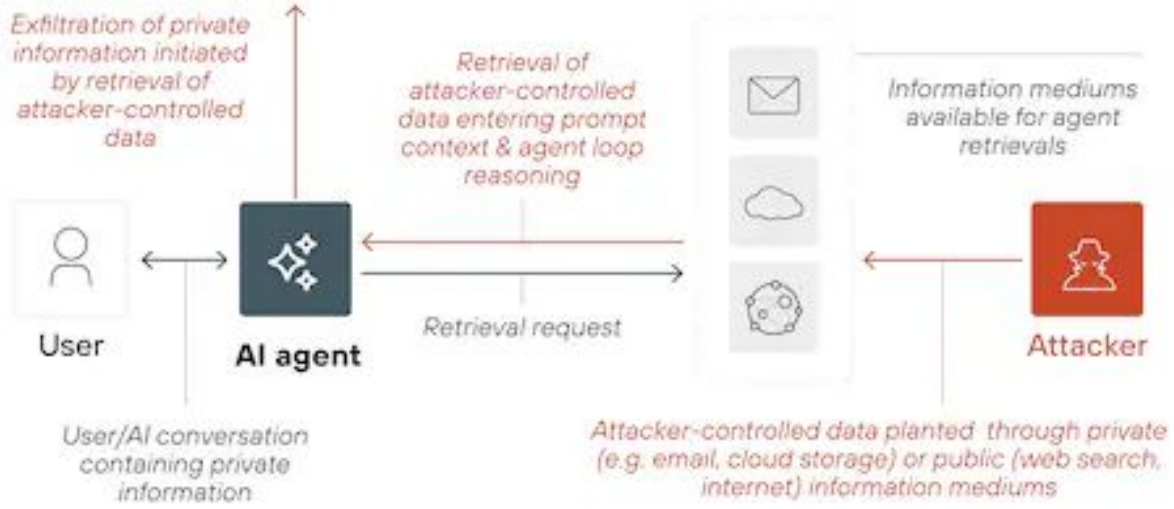
LLM need to listen your voices using role play, context layers, task layer, and constraint layers

Other popular framework:

- **With MVC (Model, View, Controller)**
- **Layered Architecture**

Layer	Function	Example Directive
Role Layer	Defines expertise or stance	"Act as a senior epidemiologist trained in peer-reviewed publishing."
Context Layer	Adds background, data source, or audience	"You are preparing a manuscript for The Lancet Global Health."
Task Layer	States the action	"Summarize the paragraph below into 150 words for the introduction section."
Constraint Layer	Sets boundaries and output expectations	"Maintain citation placeholders and academic tone; do not invent data."

Security & Code Quality Prompts



An attacker using indirect prompt injection to exfiltrate sensitive data

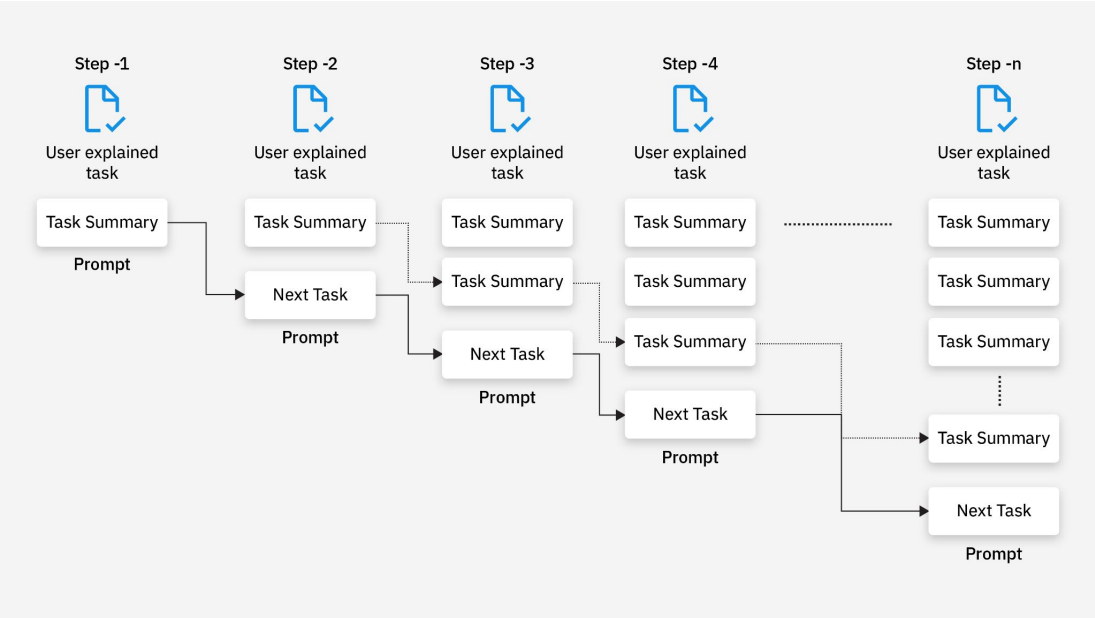
NOTE: RISKKKK !!!!

Prompt Chaining for Complex Logic

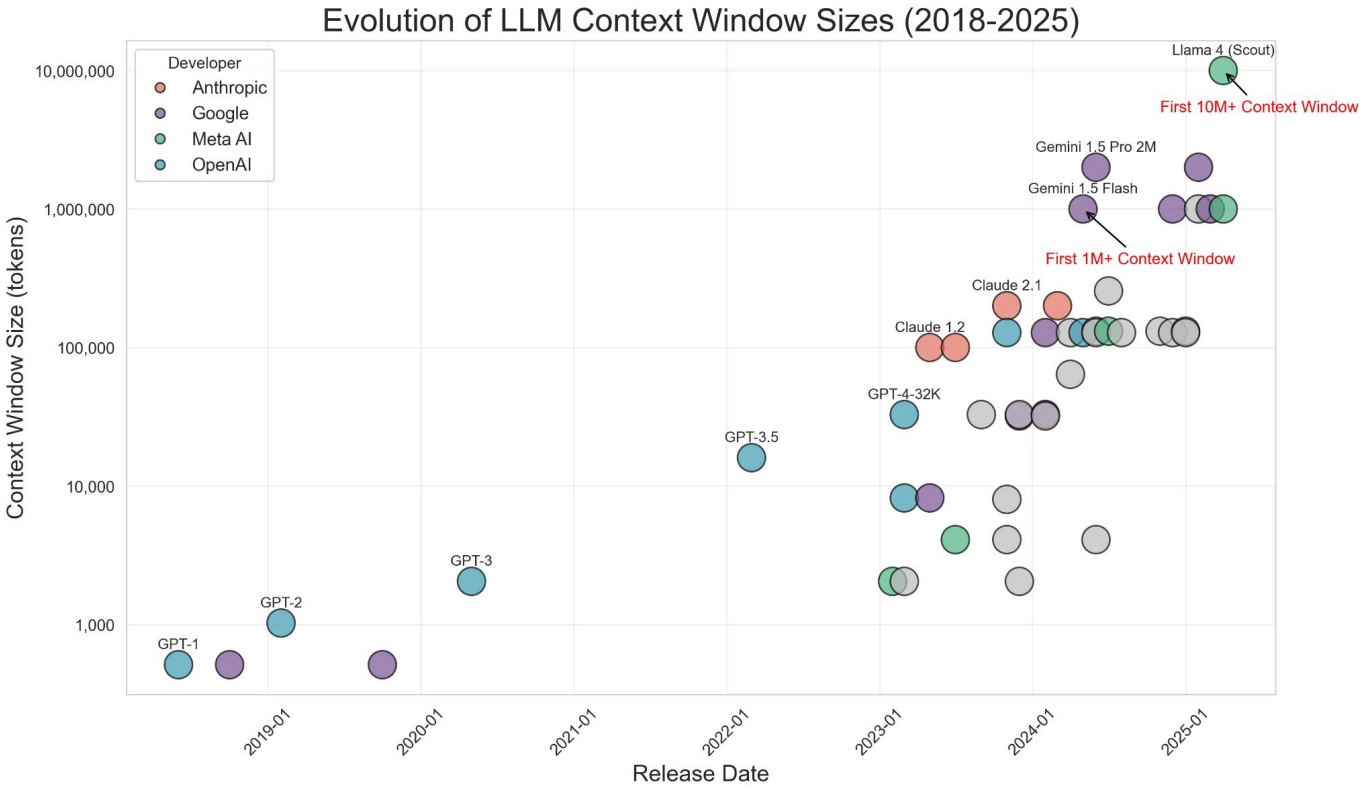
Simple Ideas like human:

More accurate > Easier debugging > Better control >> Better output

- Step 1: “Extract key facts from this text”
- Step 2: “Turn those facts into a summary”
- Step 3: “Convert summary into presentation slides”



Context Management for Better Results



NOTE: Less noise = better output. HIGHLIGHT IN 2025

Advanced Copilot Features

JABIL VTI Academy

Agents Orchestration

- Custom AI assistants for specific tasks
- Automate workflows and multi-step actions

Skills Capabilities

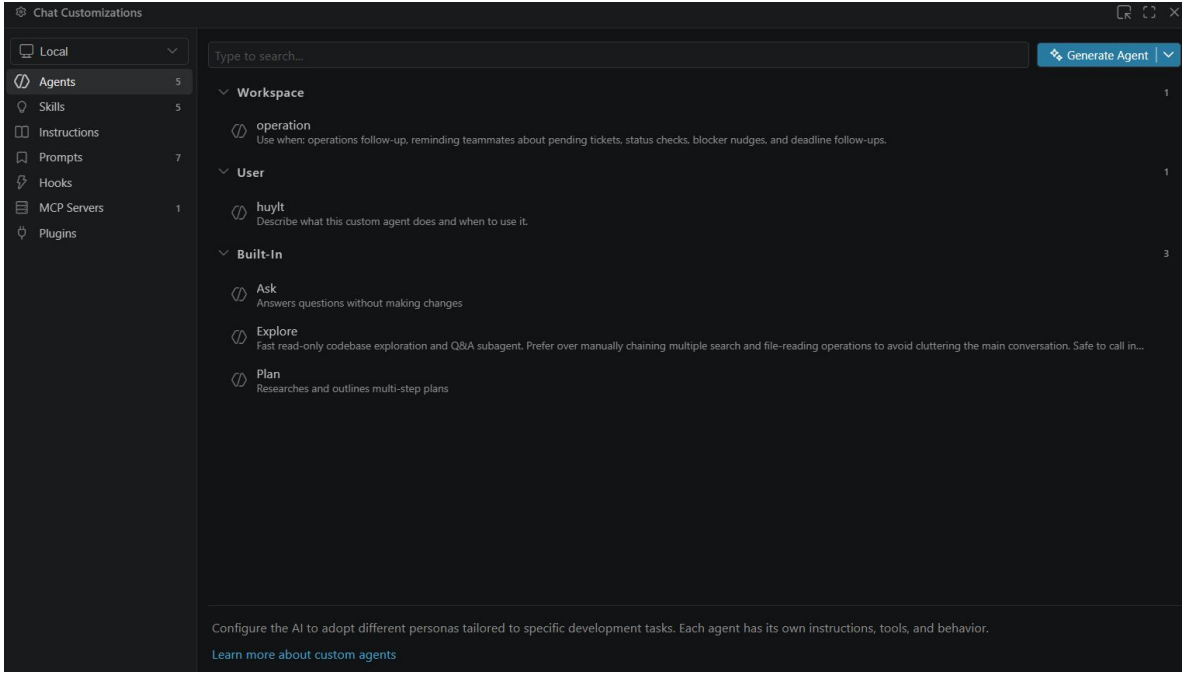
- Reusable capabilities (e.g., code, search, analysis)
- Extend agent functionality

MCP (Model Context Protocol)

- Reusable capabilities (e.g., code, search, analysis)
- Extend agent functionality

Instructions / Prompts / Plugins

- Define behavior, rules, and context
- Guide how the AI responds
- Predefined inputs for common tasks
- Ensure consistency and speed
- Predefine Packages



Advanced Copilot Features - Agents

Who are you? What are your roles

Create a simple agent in VS Code using GitHub Copilot.

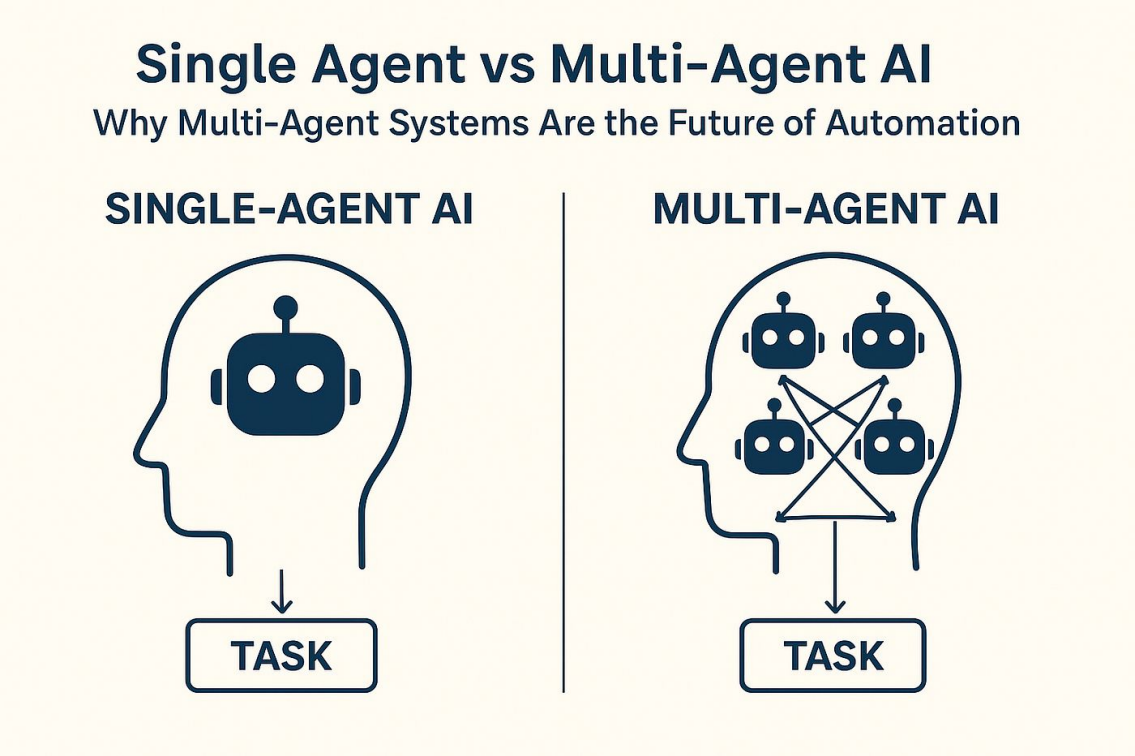
Scenario: Stock & News Agents

- Build a Stock Agent
- Build a News Agent

Example:Date: April 10

- News Agent reports: “Iran x US news todays”
    - Stock Agents reports “Buy or Not”

Hint: /create-agent in Chat Copilot



Advanced Copilot Features - Skills

Skills - Capabilities that agents can do

Create a simple agent in VS Code using GitHub Copilot.

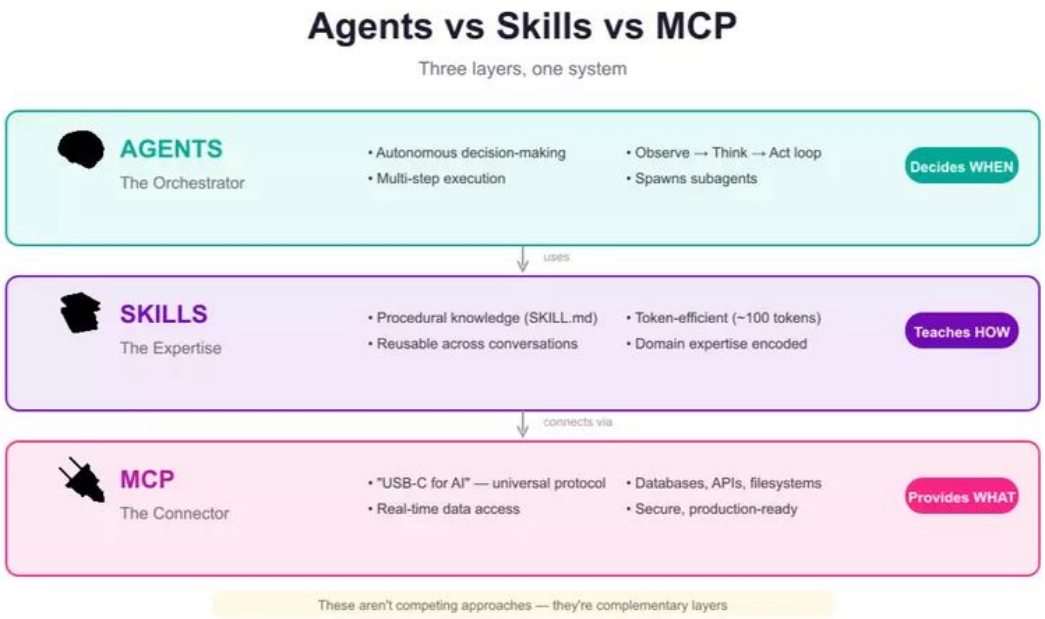
Scenario: Stock & News Agents

- News Agent: Access to URL news and stock financial index VN Index

Example

- News Agent can access to Internet: “Latest news reports high risk of wars between Iran x US today.”

Hint: /create-skills in Chat Copilot



Advanced Copilot Features - MCP

MCP - External connection

Create a simple MCP in VS Code using GitHub Copilot.

Scenario

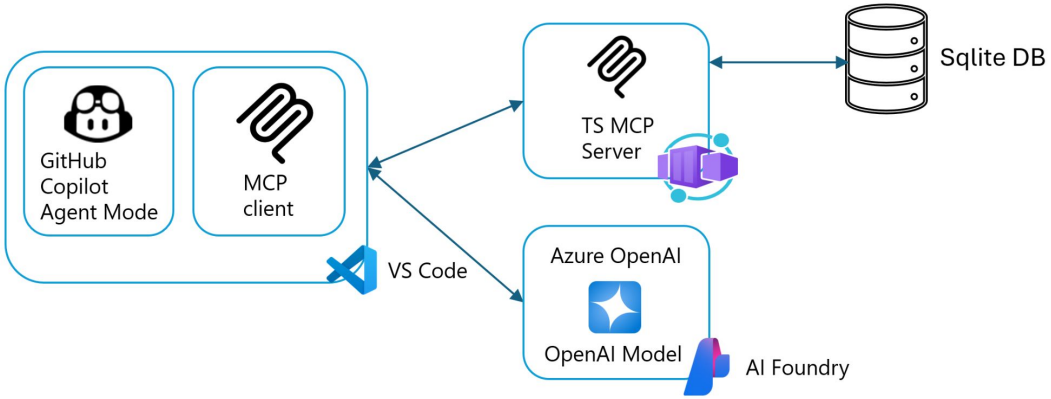
Build an agent that can use Notion services

Example

- After Stock Agents analysis about stock and send report to me
- And make report in daily

Other:

- Supabase MCP
- Chroma DB MCP
- …



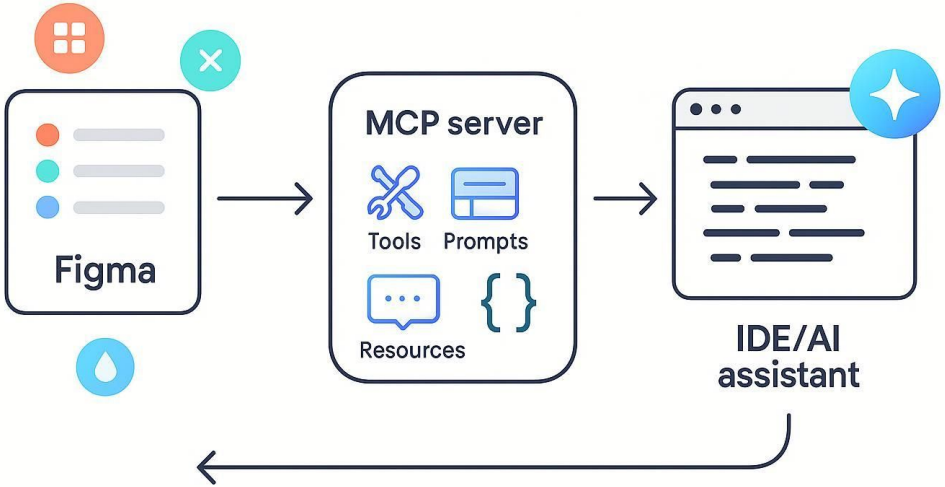
Advanced Copilot Features - Exercise #1

Design Agent with Figma MCP

Goal: Research ideas and turn them into UI drafts

- **Agent:** Understands design request
- **Tools:** Internet search for inspiration, trends, references
- **MCP:** Figma to create frames, components, layouts
- **Skill:** Standard flow for research → wireframe → polished draft

👉 Output: Figma design generated from web research



Advanced Copilot Features - Exercise #2

Data Agent with Supabase MCP

Goal: Turn local CSV data into a queryable database

- **Agent:** Understands data and user questions
- **Tools:** CSV parsing, cleaning, SQL generation
- **MCP:** Supabase to ingest local CSV into tables
- **Skill:** Upload → schema detect → load → query with SQL

👉 Output: CSV stored in Supabase, ready for SQL analysis

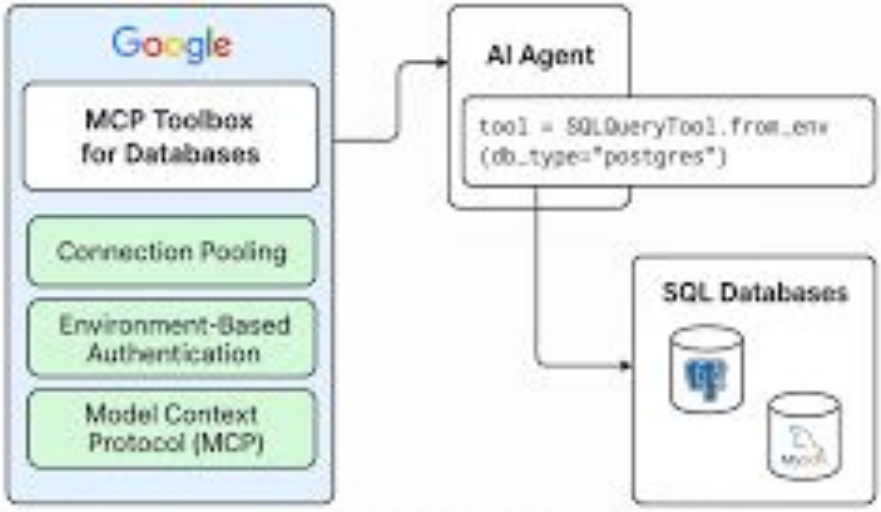


Image Source: Marktechpost.com

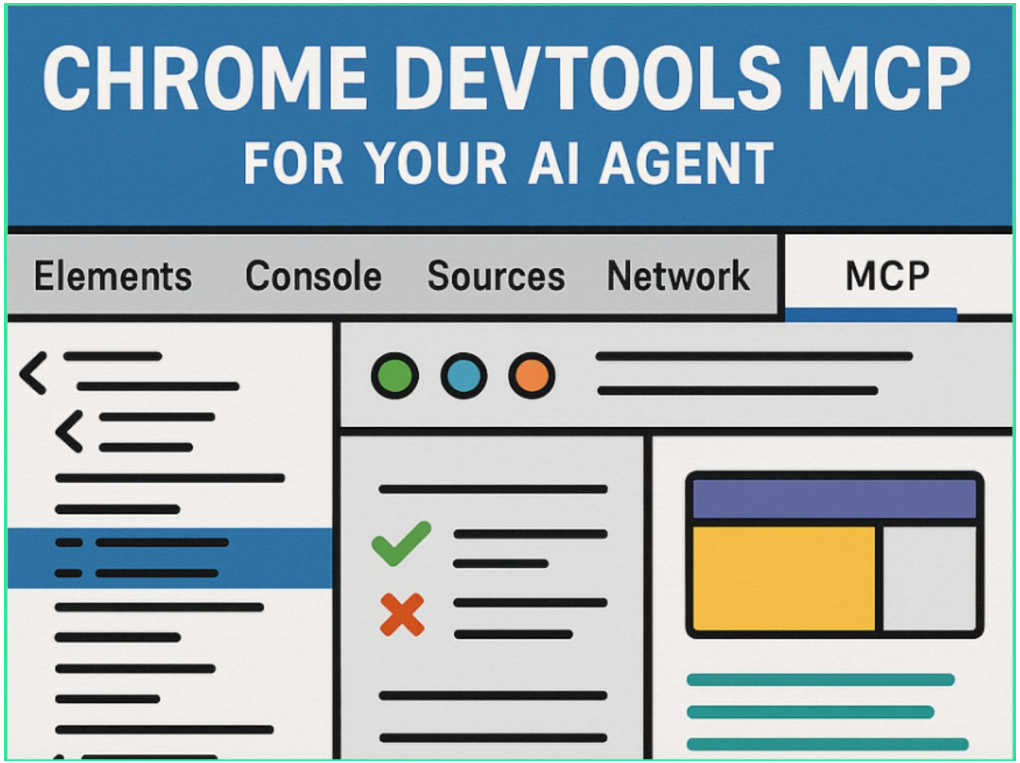
Advanced Copilot Features - Exercise #3

QA Agent with Chrome Dev MCP

Goal: Automate browser testing like Selenium

- **Agent:** Runs test scenarios and checks results
- **Tools:** Browser automation, screenshot, DOM inspection
- **MCP:** Chrome Dev tools for end-to-end test execution
- **Skill:** Open page → run steps → validate UI → report bugs

👉 Output: Automated UI test report with failures and evidence



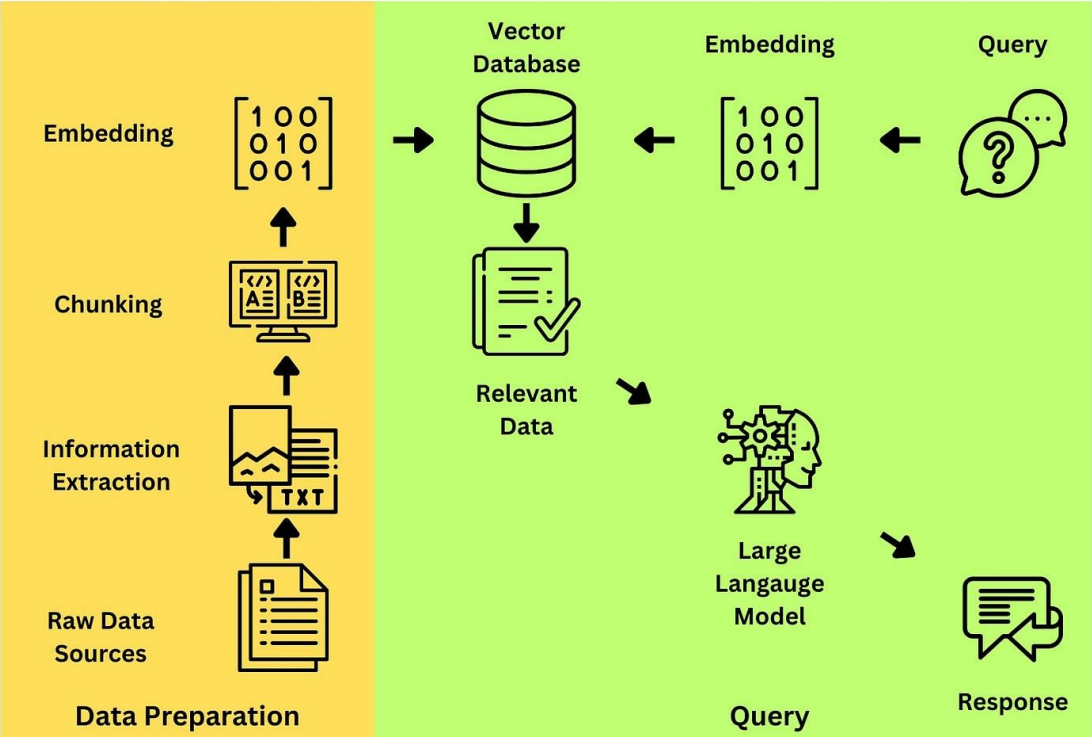
Advanced Copilot Features - Exercise #4

Knowledge Agent with ChromaDB MCP

Goal: Build local document search and retrieval

- **Agent:** Answers questions from private documents
- **Tools:** Document parsing, chunking, embeddings, retrieval
- **MCP:** ChromaDB to ingest and store local docs
- **Skill:** Import docs → index → search → verify answers

👉 Output: Searchable local knowledge base with document checking



Building Application Session 2
Pet Work Shop

Pets workshop

This repository contains the project for three guided workshops to explore various GitHub features. The project is a website for a fictional dog shelter, with a Flask backend using SQLAlchemy and Astro frontend using Svelte for dynamic pages.

Ready to get started? Let's go! The workshop scenario imagines you as a developer volunteering your time for a pet adoption center. You've been asked to add a filter to the website to allow people to limit their search results by breed and adoption status. You'll work over the next 5 exercises to perform the tasks!

1. Clone the repository and start the app for the workshop.
2. Add an endpoint to the server to list all breeds.
3. Explore the project to get a better understanding of what needs to be done.
4. Create custom instructions to ensure Copilot chat has additional context.
5. Add the new feature to the website, and ensure it works!

Build Application AI - YOUR TURN

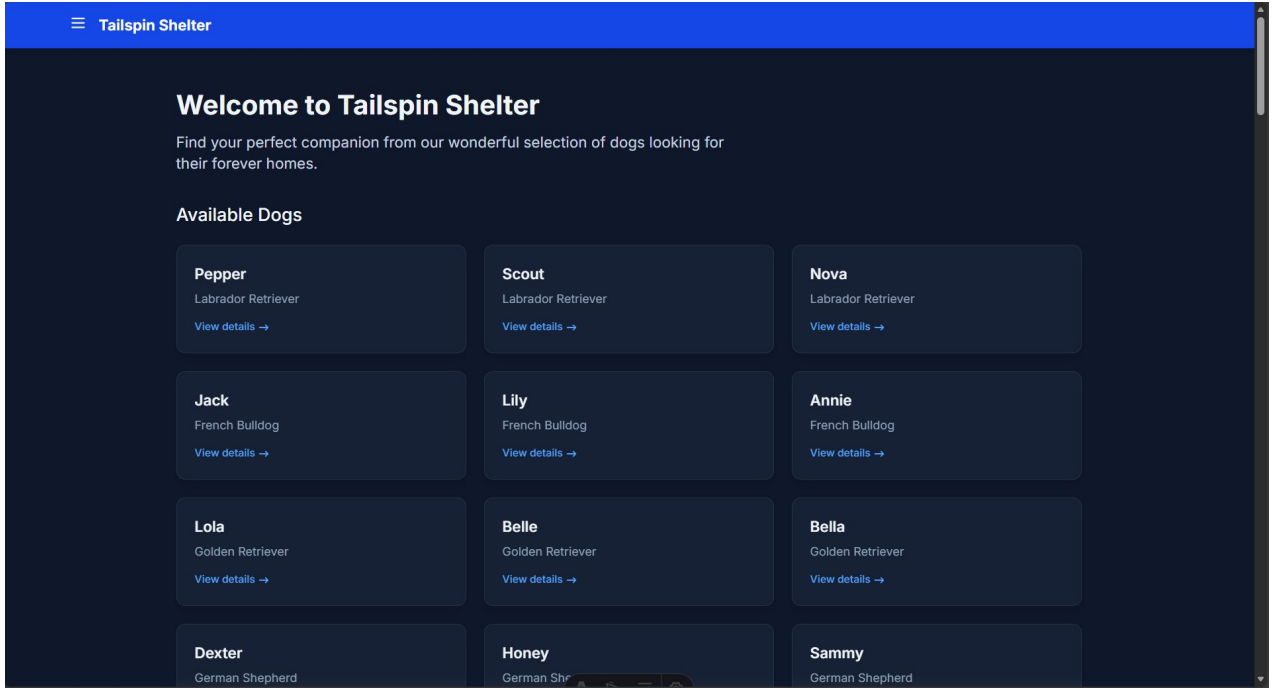
1. Fork it Repo [GitHub - meowz123/session_2_pet_workshop · GitHub]

2. Ask Github Copilot overview projects

3. Open pets-workshop /content/1-hour

4. User requirements want to have search button on right top corner

5. User requirements want to images dogs into UI easy for them to run biz



Jabil Chat Bot UI

Build Application AI - Jabil Web UI

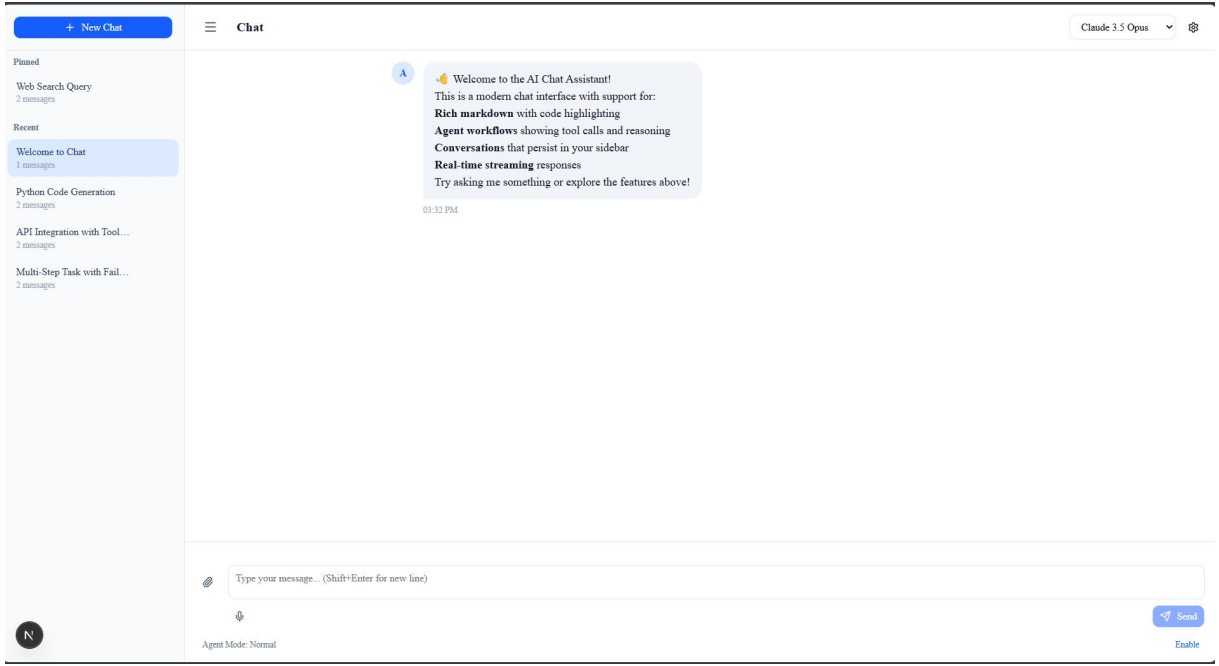
Prompt It !!!

Functional:

- Chat with AI
- Send / receive messages
- View chat history

Advanced:

- Login / logout
- Save chats (DB)
- File upload + Q&A
- Model selection
- UI: dark/light mode



Recap Summary
Exercise Next Session

Github Copilot - Recap Session 2

JABIL VTI Academy

Advanced Github Copilot

🧩 Advanced Prompt Engineering

💻 Advacned Github Coplilot

🌿 Agent Exercise

Petwork Shop

🤖 Fork Repo Do one hour

⚡ Understand full CI/CD with GitHub

Jabil ChatBot UI

🚀 Simple AI chat app

✍️ Use Copilot to generate code

🐞 Run, test, and debug

Github Copilot - Exercise Session 3

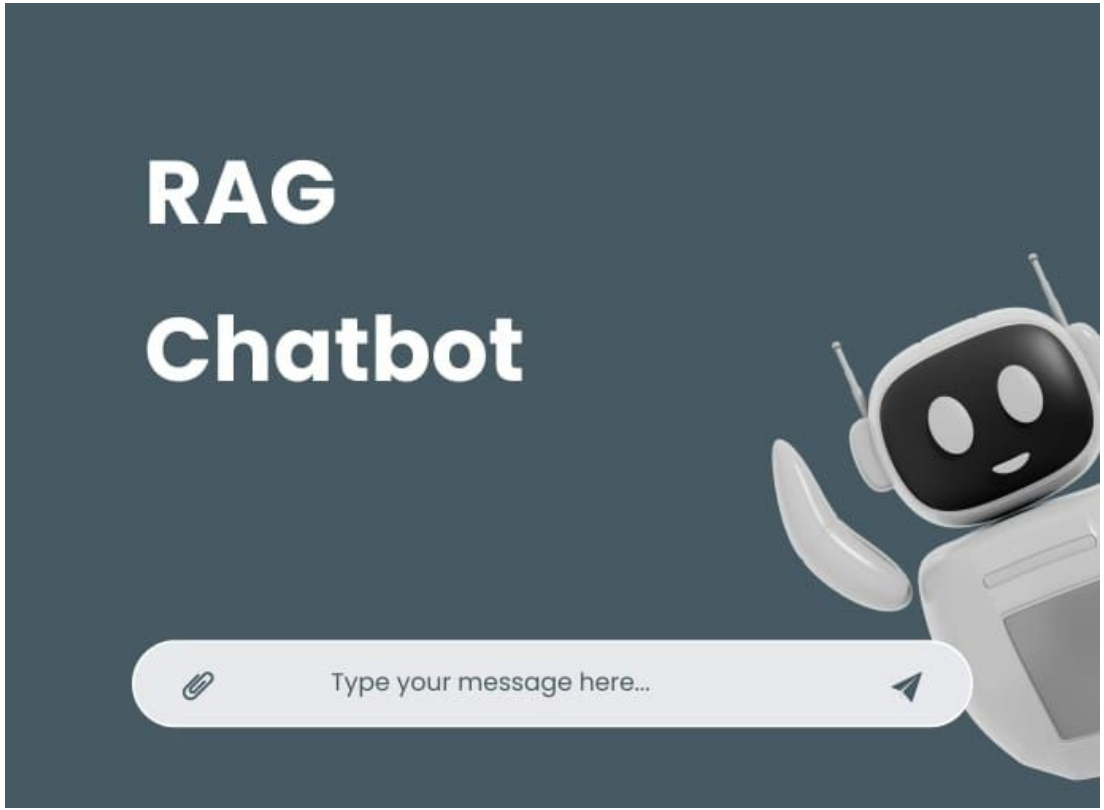
Prompt It !!!

Functional:

- Chat with AI
- Send / receive messages
- View chat history

Advanced:

- Vector DB
- RAG model



Thank you